Domino Logic Implemented for Sub-Threshold and Near-Threshold Operation

Design and Simulation of an 8-Input OR Gate in Cadence Virtuoso

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Abstract

This project explores the design and implementation of ultra-low power, high-speed digital circuits using domino logic operated in the sub-threshold and near-threshold CMOS regions. Beginning with a foundational CMOS inverter study and progressing through 4-input and 8-input OR gate implementations, the work evaluates power, delay, and switching-threshold trade-offs across the 180nm and 90nm technology nodes using Cadence Virtuoso. The final 8-input domino OR gate, simulated at Vdc = 0.35V in 180nm technology, achieved an average power of 3.377 nW with delay figures of 94.16 ns and 290.2 ns, confirming that wide fan-in domino logic is feasible in the sub-threshold regime for ultra-low-power applications such as IoT devices, biomedical implants, and portable electronics.

1. Objective

The objective of this work is to design and implement ultra-low power, high-speed digital circuits by leveraging domino logic in sub-threshold and near-threshold CMOS technology, demonstrated through the progressive realization of a 4-input and finally an 8-input OR gate in Cadence Virtuoso. This approach targets reduced energy consumption in advanced VLSI systems while maintaining robust performance for practical microelectronics applications.

2. Technology and Tools

2.1 Technology Used

- CMOS technology at 180nm and 90nm nodes, supporting sub-threshold and near-threshold operation for minimized leakage and dynamic power.
- Domino logic family: dynamic CMOS logic featuring precharge/evaluate phases with static CMOS inverters at the output, enabling rail-to-rail logic swings and cascading stages.
- Adaptive techniques such as Dual-Vth, Foot Domino, and MTCMOS considered for further leakage control and power optimization.

2.2 Tools Used

- Cadence Virtuoso — schematic entry, transient/DC simulation via the Analog Design Environment, and layout verification (DRC, LVS).
- gpdk180 and gpdk90 nm technology libraries (PDKs).

3. Literature Review

Sharma et al. present domino logic as an advanced CMOS dynamic-logic technique that speeds up circuits while enabling rail-to-rail logic swing, overcoming static CMOS limitations via a precharge/evaluate phase with an output static inverter. Their implementation of AND, OR, NAND, and NOR domino gates showed reduced transistor count, smaller silicon area, and lower power consumption relative to static CMOS.

Verma et al. describe domino logic as an evolution of dynamic logic that inserts a static inverter between cascaded stages to ensure signal stability and support large fan-in designs. While this adds a PMOS inverter, the approach balances speed and noise margin, offering smaller area and faster operation than static CMOS at the cost of added clocking complexity.

Crupi et al. examine the impact of high-mobility channel materials (e.g., SiGe pMOSFETs) on near-threshold and sub-threshold CMOS logic. Their results show high-mobility materials meaningfully improve the energy-performance trade-off in near-threshold circuits, but in the sub-threshold regime the same gains can be achieved simply by tuning the threshold voltage of conventional silicon devices — making channel-material innovation less relevant below threshold.

Ankur Kumar et al. propose a bulk-driven keeper transistor for wide fan-in domino OR circuits (up to 64-bit fan-in) to reduce delay, power variation, and overall dissipation. Verified at 45nm under voltage, temperature, and process variation using Cadence Virtuoso/SPECTRE, the technique achieved a 31% reduction in power variation, 25% reduction in delay variation, and a 1.56× improvement in noise immunity over conventional domino circuits.

Sanapala et al. explore CMOS logic families in the sub-threshold region for ultra-low-power applications, providing broader context for the family-level trade-offs this project builds on.

4. Design Progression

4.1 Review 1 — Proposal and Work Plan

The initial phase established the work plan: literature review of domino logic structures for sub-threshold/near-threshold operation; schematic design of a 4-input OR gate using dynamic logic with an output inverter for signal restoration in 180nm technology; transient simulation benchmarked against static CMOS; adaptation of the circuit for sub-threshold and near-threshold conditions; and final performance analysis focused on power, speed, area, and leakage.

4.2 Review 2 — CMOS Inverter Characterization and the 4-Input OR Gate

Before scaling to wide fan-in domino gates, a CMOS inverter was characterized in both the normal operating region and the sub-threshold region, to establish a baseline for switching threshold, delay, and power.

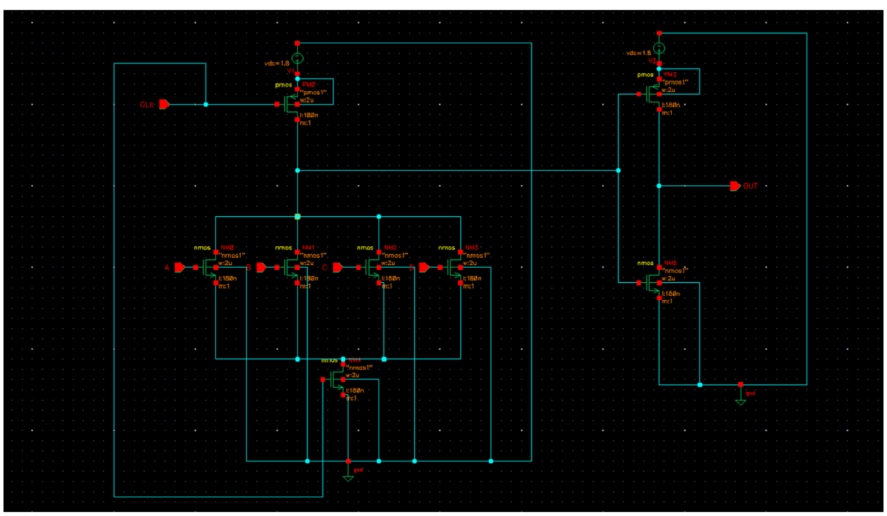

Figure 1. CMOS inverter schematic — normal operating region (Vdc = 1.8V, 180nm).

At Vdc = 1.8V, the inverter showed a propagation delay of 1.334 ps and an average power of 1.894 µW. The inverter was then re-simulated in the sub-threshold region at Vdc = 0.35V, first without transistor sizing (pwidth = 2µ, nwidth = 2µ) and then with sizing (pwidth = 6µ, nwidth = 4µ), and finally across the gpdk180 and gpdk90 technology libraries.

Sizing	Switching Threshold (mV)
pwidth = 2µ, nwidth = 2µ	127.1969
pwidth = 6µ, nwidth = 4µ	132.20

Technology	Average Power (W)	Delay (s)
180nm	3.971×10^{-9}	265×10^{-12}
90nm	6.204×10^{-9}	111.8×10^{-12}

The 4-input OR gate was then implemented using the domino logic topology (precharge PMOS, evaluate NMOS stack, and a static output inverter), and its functional correctness was verified across two independent input transient test sets.

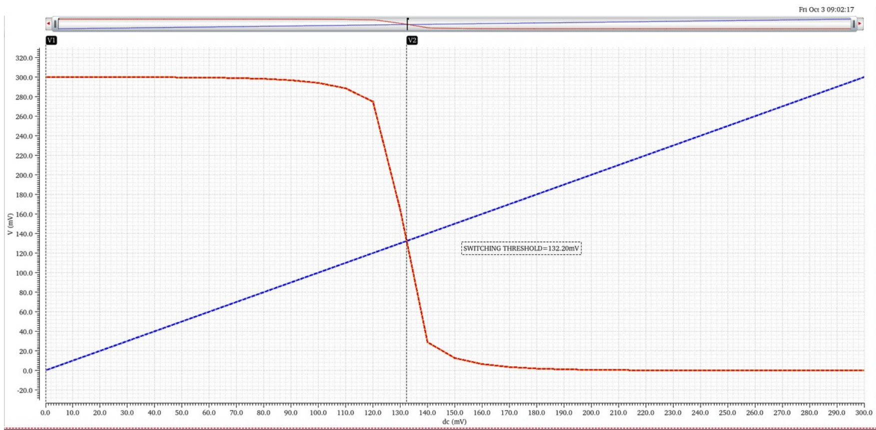

Figure 2. 4-input domino OR gate schematic implemented using gpdk90 technology.

4.3 Review 3 — Scaling to an 8-Input OR Gate

Building on the validated 4-input design, the final phase scaled the domino OR gate to 8 inputs at Vdc = 0.35V in 180nm technology, keeping the same precharge/evaluate/keeper structure.

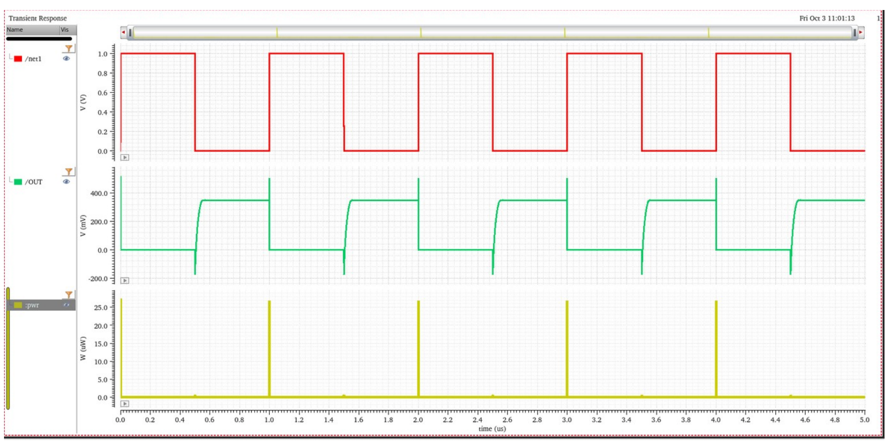

Figure 3. 8-input domino OR gate schematic (sub-threshold, 180nm).

Transient simulation confirmed correct rail-to-rail OR behavior across all 8 inputs, with the precharge (CLK) and evaluate phases clearly visible on the output and power waveforms.

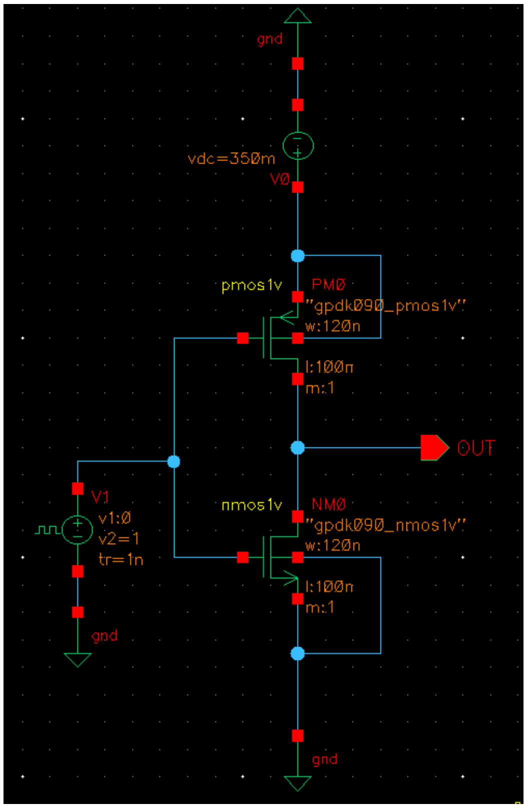

Figure 4. Transient response of the 8-input domino OR gate showing CLK, inputs A–H, OUT, and instantaneous power.

5. Results

5.1 Sizing and Threshold Voltage — Review 3

Sizing	Switching Threshold (mV)	Threshold Voltage (V)
pwidth = 2μ, nwidth = 2μ	127.1969	Pmos = -0.473, Nmos = -0.518
pwidth = 6μ, nwidth = 4μ	132.20	Pmos = -0.48, Nmos = 0.525

5.2 Power and Delay — Final 8-Input OR Gate

Technology	Average Power (W)	Delay 1 (s)	Delay 2 (s)
180nm	3.377×10^{-9}	94.16×10^{-9}	290.2×10^{-9}

6. Inference and Discussion

- Wide fan-in domino logic is feasible in sub-threshold mode and is well suited to low-power, moderate-speed applications.
- Compared to static CMOS, domino logic offers competitive power benefits at the cost of higher delay — the expected trade-off in this operating regime.
- Results align with literature: adaptive sub-threshold voltage control in domino logic reduces power dissipation and improves noise immunity, particularly for wide fan-in gates.

- 180nm is generally preferable to 90nm for sub-threshold circuit design in this study — it shows lower leakage current, stronger noise margins, and reduced sensitivity to process variation and short-channel effects, at the cost of raw speed versus 90nm.
- The sub-threshold domino approach is validated as a robust technique for ultra-low-power VLSI design, especially where energy efficiency matters more than raw speed — e.g. IoT sensors, biomedical implants, and portable electronics.

7. Conclusion

This project demonstrated the feasibility of implementing wide fan-in (up to 8-input) domino OR logic in the sub-threshold and near-threshold CMOS regions using Cadence Virtuoso. Through a staged design process — CMOS inverter characterization, 4-input OR gate validation, and final 8-input OR gate implementation — the study confirmed that sub-threshold domino logic achieves nanowatt-level average power at the cost of increased propagation delay, and that the 180nm node offers a better overall balance of reliability and power efficiency than 90nm for this class of ultra-low-power design.

References

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